Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000

COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: Bachelor of Science in Mining Engineering

Primary Job Roles students are prepared for in the program	**Highly competitive and conscientious mine managers and mine planners capable of managing global and multicultural environment.**
Secondary Job Roles students are prepared for in the program	**Safety Engineers, Explosive Engineers, Environmental Engineers, Government Engineers, Shift Production Engineers, Shift Production Engineers, Exploration Engineers, Academicians, Grade Control Engineers, Division Managers, Community Relations Engineers**

Primary and Secondary Job Roles provide a mental anchor to the identification of program outcomes you will be having for your program. This will allow you to check for completeness in your program outcomes.

Current CMO Guiding Program:

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of policies, standards and guidelines in the establishment of an outcome-based education (OBE) system in higher education institutions offering engineering programs as advocated under CMO 37 s. 2012 and policy-standard to enhance quality assurance (QA) in Philippine higher education through an outcomes-based QA under CMO 46 s. 2012, and by virtue of Commission en banc Resolution No. 788-2017 dated October 24, 2017 the following policies and standards are hereby adopted ad promulgated by the Commission.

This CMO integrates the establishment of an outcomes-based Based on the Guidelines for the Implementation of CMO No. 46 series of 2012 and CMO 37 s. 2012, this PSG implements the shift to outcomes-based education (OBE) leading to competency-based standards. It specifies the "core competencies" expected of Bachelor of Mining (BSEM) graduates "regardless of the type of Higher Education Institutions (HEI) they graduate from." However, in recognition of outcomes-based education and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

Program Description:

Definition
Mining Engineering is a profession that deals with the application of mathematics, natural and applied sciences, humanities and social sciences in mineral deposit assessment, mine feasibility study, mine design, development and construction, management of mining operations, mining and mineral economics, environmental protection and rehabilitation, mine safety, closure and decommissioning to harness the mineral resources.

Degree Name
In consonance with the field of specializations of the Mining Engineering Profession stipulated in section 5, article 4 of RA 4274 also known as Mining Engineering Law, the degree program herein shall be called BACHELOR OF SCIENCE IN MINING ENGINEERING (BSEM).

Nature & Scope of the Practice of an EM Graduate
The scope of the practice of Mining Engineer is defined in the Mining Engineering Law of Republic Act No. 4274 and pertains to professional service in Mining Engineering in the form of consultation, investigation, management, valuation, planning, capacity building and education, designing or supervision of operation. Mining Engineering embraces to: (a) Management, engineering or supervisory positions in mines, mine installation or construction projects where knowledge in mining engineering is necessary in carrying out such function; (b) Calculation, estimation and certification of mineral and/or ore reserves, preparation of mine pre-feasibility and feasibility studies; mine valuation, mine auditing, mine consulting, third party independent opinion, third party mine audits and appraisal of mining machinery and equipment; (c) participation in the preparation of environmental studies for mining projects and monitoring under the Environmental Impact Assessment (EIA) system that calls for knowledge of Mining Engineering; (d) Preparation, approval or signing of mining documents such as but not limited to mine reports, mine plans and designs, specifications or project estimates, and mining engineer's report; (e) Design of mines, mining methods and applicable machinery, works or installation and commissioning or decommissioning of mines; (f) Employment in government service requiring professional knowledge of mining engineering; (g) Management or supervision of small-scale mines or similar projects as prescribed by law; (h) Formulation of Mineral Reporting Codes and/or Engineering/Technical standards for use by Professional and Registered Mining Engineers; and, (i)Teaching, lecturing and reviewing of professional mining engineering subjects in government recognized and accredited universities, colleges, schools and institutes.

VISION

A PREMIER UNIVERSITY IN THE ASEAN REGION
By becoming a premier university in the ASEAN Region, the USEP shall be a center of excellence and development, responsive and adaptive to fast changing environments. USEP shall be known as the leading university in the country that fosters innovation and applies knowledge to create value towards social, economic, and technological developments.

MISSION

USeP shall produce world –class graduates and relevant research and extension through quality education and sustainable resource management.
Particularly, USeP is committed to:
1. Provide quality education for students to grow in knowledge, promote their well-rounded development, and make them globally competitive in the world of work;
2. Engage in high impact research, not only for knowledge's sake, but also for its practical benefits to society; and,
3. Promote entrepreneurship and industry collaboration.

CORE VALUES

USeP is a community of scholars that values:
U – Unity
S – Stewardship
E – Excellence
P – Professionalism

	Program Educational Objectives	Mission-Vision and Core Value Alignment								Reference Source or CMO Provision Compliance
		Vision	Mission			Core Values				
			1	2	3	U	S	E	P	
PEO1	Establish expertise in the field of Mining Engineering.	✓	✓				✓	✓	✓	USEP Brochure, CMO 101
PEO2	Actively participate in professional communities to nourish ever-developing careers	✓		✓		✓	✓		✓	USEP Brochure, CMO 101
PEO3	Demonstrate leadership and commitment to work in diverse cultural and interdisciplinary backgrounds.	✓	✓	✓		✓		✓	✓	USEP Brochure, CMO 101
PE04	Strive to be globally competitive.	✓	✓		✓		✓	✓	✓	USEP Brochure, CMO 101

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting statements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation

Program Attributes:

PG01 – Critical Analysis
PG02 – Leadership and Teamwork
PG03 – Service-Oriented
PG04 – Professional Competence
PG05 – Communication
PG06 – Life-Long Learning

Program Attribute		Program Outcomes	Performance Indicators
PGO1	a	Apply knowledge of mathematics and sciences to solve engineering problems;	Apply concepts in mathematics to solve engineering problems.
PGO4			Apply principles in chemical sciences to solve engineering problems.
PGO6			Apply principles in physical science to solve engineering problems.
PGO1, PGO4	b	Design and conduct experiments as well as to analyze and interpret data	Perform laboratory experiments as prescribed by standard procedures.
PGO5, PGO6			Design an experimental to achieve a specific objective.
PGO1, PG02	c	Design a system, component, or process to meet desired needs within realistic constraints such as economic, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	Develop an engineering solution that have various constraints as factors.
PG03, PG04			Design a system or process based on various constraints.
PG05, PG06			
PG02	d	Function on multidisciplinary teams	Shows tact in dealing with people he/she come in contact with.
PG03			Capable of learning from and listening to co-members.

PG06			Shows respect and courtesy in dealing with peers and superiors.
PG01	e	Identity, formulate, and solve engineering problems	Interpret statement problems into mathematical expressions.
PG04			Apply concepts, theories, and principles in solving engineering problem(s).
PG06			Solve specific engineering problem related to discipline.
PG04	f	Understand professional, social and ethical responsibility	Internalizes the professional code of ethics for specific discipline.
PG05			Applies the principle of professional ethical standard and social responsibility to actual practice.
PG06			
PG04	g	Communicate effectively	Articulate the convention and style in oral communication that conveys complete thought and information.
PG05			Articulate the convention and style in oral communication that conveys complete thought and information.
PG06			Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
PG01, PG03	h	Understand the impact of engineering solutions in a global economic, environmental, & societal context through broad education	Relate engineering solutions to global, economic, environmental, and societal context.
PG04, PG06			Describe engineering solutions in application to global, economic, environmental, and societal context.
PG01, PG02	i	Recognize the need for, and an ability to engage in life-long learning	Demonstrate ability to learn independently.
PG03, PG04			Shows appreciation and understanding of the engineering profession.
PG05, PG06			Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
PG02, PG03	j	Apply knowledge of contemporary issues	Identify the socio-economic, political, and environmental issues related to the practice of engineering.
PG04, PG05			Explain the possible impact of technology to local, national, and international issues.
PG06			Shows leadership skills in managing a multidisciplinary group.
PG01	k	Use technique, skills, and modern engineering tools necessary for engineering practice	Handles the details of the work assigned to him/her.
PG04			Demonstrate the ability to operate the equipment needed in the engineering practice.
PG06			Use modern technology in performing engineering practice.
PG01, PG02	l	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	Capable of learning from and listening to co-members.
PG03, PG04			Shows leadership skills in managing a multidisciplinary group.
PG05, PG06			Participates in discussions and comes up with sound suggestions to problems.

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

List courses such that first year first semester courses appear first, until the last semester of the last year of the degree program. Exclude in the list the following: (a) General Education courses (b) Elective courses. As much as possible, observe the following:

a. *For every course of 3-units, do not exceed 3-4 Graduate Outcomes you will be focusing and targeting to develop in the course*
b. *For every course of 1-units, do not exceed 1-2 Graduate Outcomes you will be focusing and targeting to develop in the course*
c. *For every course of 5-units, do not exceed 4-5 Graduate Outcomes you will be focusing and targeting to develop in the course*
d. *Ensure that all the Graduate Outcomes have courses developing it for every year level*
e. *Total the number of X's horizontally per course and vertically per Graduate Outcome*

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

Program Outcomes	
Upon completion of the program, the University of Southeastern Philippines Mining Engineering students will be able to:	
a	Apply knowledge of mathematics and sciences to solve engineering problems;
b	Design and conduct experiments, as well as to analyze and interpret data
c	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.
d	Function on multidisciplinary teams
e	Identify, formulate, and solve engineering problems
f	Understand professional, social and ethical responsibility
g	Communicate effectively
h	Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education
i	Recognize the need for, and an ability to engage in life-long learning
j	Apply knowledge of contemporary issues
k	Use techniques, skills, and modern engineering tools necessary for engineering practice
l	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments
m	Understand at least one specialized field of mining engineering practice

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes											
				a	b	c	d	e	f	g	h	i	j	k	l
FIRST YEAR	1ST SEM	3	Purposive Communication												
		3	Mathematics in the Modern World												
		3	Engineering Calculus 1	I				I							
		4	Chemistry for Engineers	I	I			I							
		1	Engineering Drawing and Plans				I			I					
		3	Principles of Geology	I										I	
		2	Movement Enhancement												
		3	National Service Training Program 1												
	TOTAL	22													
	2ND SEM	3	Understanding the Self												
		3	Ethics												
		3	Science, Technology and Society												
		3	Engineering Calculus 2	E				E							
		4	Physics for Engineers	E	I		I	E						I	
		1	Computer Aided Drafting						I	I		I		I	
		3	Elementary Mineralogy	I								I			
		2	Fitness Exercise												
		3	National Service Training Program 2												
	TOTAL	25													
OFF SEMESTER 1		3	Differential Equations	E				E							
		3	Petrology	E										E	
		3	Principles of Mining	I								I			
	TOTAL	9													
SECOND YEAR	1ST SEM	3	Readings in the Philippine History												
		3	General Education Elective 1												
		3	The Life and Works of Rizal												
		3	Engineering Data Analysis	I				I			I				
		3	Underground Mining	E		E		E	E		E				
		3	Principles of Metallurgy	E								E			
		3	Structural Geology	E	E			E		E				E	
		2	Physical Activities Towards Health & Fitness I												
	TOTAL	23													
	2ND SEM	3	Arts Appreciation												
		3	Surface Mining	E		E		E	E		E				
		3	Mineral Deposits	E						E	E				
		3	Mineral Processing	E		E		E				E			
		2	Techniques of Metallurgical Analysis	E	E					E				E	
		3	General Surveying 1	E	E									E	
		2	Physical Activities Towards Health & Fitness II												
	TOTAL	22													

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes											
				a	b	c	d	e	f	g	h	i	j	k	l
THIRD YEAR		2	Techniques of Mining Analysis	E							E	E		E	E
		3	Mine Safety	E		E		E	E	E	E	E	E	E	
		3	Mine Surveying	E							E	E		E	E
		3	Engineering Mechanics	E											
		3	Fluid Mechanics	E				E							
		3	Environmental Science and Engineering						E		E		E		E
		3	Contemporary World												
		3	General Education Elective 2												
	TOTAL	23													
	2ND SEM	3	General Education Elective 3												
		3	Mine Ventilation	E	E	E		E	E		E	E		E	
		2	Basic Mechanical Equipment							E	E	E			
		3	Basic Electrical Engineering	E	E			E							
		3	Track Specialization 1	D							E	E		E	E
		4	Mechanics of Deformable Bodies	E				E							
		3	Engineering Economics	E					I		I		I		I
		3	Methods of Research						E	E	E		E		E
	TOTAL	25													
OFF SEMESTER 2		2	Mine and Plant Practice (320 Hrs)				D		D	D		D	D	D	
	TOTAL	2													
FOURTH YEAR	1ST SEM	3	Mine Research and Study/Thesis		D	D		D		D	D	D		D	
		3	Mine Design	D	D	D		D	D	D	D	D	D	D	
		3	Mine Environmental Management	E		E	E	E	E	E	E	E			
		3	Mining and Environmental Laws and Ethics				E	E	D	E		E	D		
		3	Track Specialization 2	D							E	E		E	E
		3	Rock Mechanics	D	E	E		E				E		E	
		3	Engineering Management				E		E	E	E		E		E
		3	Technopreneurship				E		E	E					E
	TOTAL	24													
	2ND SEM	3	Mine Research and Study	E		D				D			E		
		1	Mining Seminar								E	E		E	D
		1	Correlation Course	D				D				D			
		3	Mine Management	D		D	E	E	E	E		E	E		
		3	Mine Economics	D				D		D					
		3	Track Specialization 3	D							E	E		E	E
		3	Energy Resources								E	E		E	E
		3	Free Elective												
	TOTAL	14													

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

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